Cybersecurity Incident Report: Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.
The analyzer logs show that when I send UDP packets to the DNS server, I receive ICMP packets containing the error message; “udp port 53 unreachable”. Port 53 is a port normally used for DNS service. This may indicate higher than normal incoming traffic, likely a malicious DDoS attack.

Part 2: Explain your analysis of the data and provide at least one cause of the incident.
The incident occurred this afternoon and customers began reporting the same issue to the IT team so we attempted to go to the website ourselves and met the same issue. The webpage wouldn’t load. We then loaded our network analyzer tool, tcpdump, and attempt to load the page again. The resulting logs revealed that port 53 was unreachable, we are continuing to investigate the root cause of the issue to determine how to restore access to our website. Next steps include checking the firewall configuration to see if port 53 is blocked and contacting the system administrator for the web server to have them check for signs of an attack. Due to the abnormally sudden high volume of requests we suspect this may be a DDoS attack.